

# CSE 574 - Computational Complexity

Zhengjie Min

September 15, 2026

### **Abstract**

This is a template for TCS (and other kind of) lecture notes.

# Contents

|          |                                              |           |
|----------|----------------------------------------------|-----------|
| <b>I</b> | <b>Introduction &amp; Background Review</b>  | <b>2</b>  |
| <b>1</b> | <b>The Beginning</b>                         | <b>3</b>  |
| 1.1      | Definition of Computational Theory . . . . . | 3         |
| 1.2      | What is Complexity About? . . . . .          | 4         |
| <b>2</b> | <b>Time and Non-deterministic Time</b>       | <b>5</b>  |
| 2.1      | Languages and Problems . . . . .             | 5         |
| 2.2      | Classes of Complexity . . . . .              | 5         |
| 2.2.1    | Deterministic Time Complexity . . . . .      | 5         |
| 2.2.2    | Non-deterministic Time Complexity . . . . .  | 5         |
| <b>3</b> | <b>Proving NP-hard Problems</b>              | <b>7</b>  |
| 3.1      | NP-hard and NP-complete . . . . .            | 7         |
| 3.2      | Cook-Levin Theorem . . . . .                 | 7         |
| <b>A</b> | <b>Algebra</b>                               | <b>10</b> |
| <b>B</b> | <b>Analysis</b>                              | <b>11</b> |

## **Part I**

# **Introduction & Background Review**

# Chapter 1

## The Beginning

The start of computational theory can be traced back to the 1930s, when mathematicians and logicians were trying to formalize the notion of computation and algorithms. The Church-Turing thesis, proposed independently by Alonzo Church and Alan Turing, laid the foundation for the study of computational complexity.

**Theorem 1.0.1 (Church-Turing Thesis).** Every effectively calculable function is computable by a Turing machine.

Then mathematicians and computer scientists began to explore the limits of computation, with such an open thesis:

**Problem 1.0.1 (Extended (Quantum) Church-Turing Thesis).** Every physically realizable model of computation can be simulated by a (quantum) Turing machine efficiently.

### 1.1 Definition of Computational Theory

**Remark 1.1.1.** Computational theory deals with how much resources we need to solve a computational problem.

Resources can be many things. For example:

- Time
- Memory / Space
- Samples Complexity (how many samples we need to distinguish between distributions)
- Communication Rounds
- Oracle Calls
- Randomness
- Power
- Quantum Entanglement
- Number of Parallel Workers
- ...

---

## 1.2 What is Complexity About?

In CS Theory, people do research in algorithms and complexity. People might lean on one side or the other, but they are closely related. In a word:

**Remark 1.2.1.** Algorithms are about the upper bound of a problem, while complexity is about the lower bound of a problem.

## Chapter 2

# Time and Non-deterministic Time

## 2.1 Languages and Problems

In computational complexity, we talk about languages as how we think about computational problems.

**Definition 2.1.1.** A **language**  $L$  is a set of strings that  $L \subseteq \Sigma^*$ , where  $\Sigma$  is a finite alphabet (often  $\{0, 1\}$  bits.)

**Example 2.1.1.** An algorithm  $\text{PRIMES}(n)$  can decide whether a number  $n$  is prime or not. The language of this problem is:

$$\text{PRIMES} = \{\langle x \rangle : x \text{ is prime}\}.$$

$\langle x \rangle$  is the encoding of  $x$  in binary.

## 2.2 Classes of Complexity

### 2.2.1 Deterministic Time Complexity

**Definition 2.2.1.**  $L \in \text{DTIME}(T(n))$  if there's a deterministic Turing machine  $M$  that decides  $L$  in time  $O(T(n))$ , where  $n$  is the size of the input.  $P := \bigcup_{c \geq 1} \text{DTIME}(n^c)$  is the class of problems that can be solved in polynomial time.

Researchers have found a polynomial time algorithm for the PRIMES problem in 2002, so  $\text{PRIMES} \in P$ .

**Example 2.2.1.**

$$\text{CONN} = \{x : x \text{ is the adjacency matrix of a connected graph}\}$$

$\text{CONN} \in P$  because we can use DFS or BFS to check connectivity in polynomial time.

### 2.2.2 Non-deterministic Time Complexity

**Definition 2.2.2.** A **non-deterministic Turing machine** is a Turing machine that can make non-deterministic choices at each step. It can be thought of as a machine that can explore multiple computation paths simultaneously, or a machine that takes different branches of operations in different universes.

For instance, assume a certain state. In this state, the transition function for a deterministic TM might be just  $(0, L, 1)$  and  $(1, R, 1)$ , while a non-deterministic TM might have  $(0, R, 1)$ ,  $(0, L, 0)$ , and  $(1, L, 1)$ . When the machine meets a 0, it may arbitrarily choose to go left or right.

**Definition 2.2.3.**  $L \in \text{NTIME}(T(n))$  if there's a non-deterministic Turing machine  $M$  that decides  $L$  in time  $O(T(n))$  (regardless of which universe.) Here 'decides' means that if  $x \in L$ , there exists a computation path that accepts  $x$ ; if  $x \notin L$ , all computation paths reject  $x$ .

**Definition 2.2.4.**  $L \in \text{co-NTIME}(T(n)) \Leftrightarrow \bar{L} \in \text{NTIME}(T(n))$ .

**Definition 2.2.5.**  $\text{NP} := \bigcup_{c \geq 1} \text{NTIME}(n^c)$  is the class of problems that can be solved in polynomial time by a non-deterministic Turing machine.

**Remark 2.2.1.** We normally consider NP to be 'verifiable' in polynomial time, using a 'certificate'  $u$ . Consider a non-deterministic TM that accepts  $x$  in a certain path. The sequence of non-deterministic choices made by the TM can be encoded as a string  $u$ . Then we can construct a deterministic TM  $M(.,.)$  that takes  $x$  and  $u$  as input, and verify whether  $x \in L$  by checking if there exists a certificate  $u$  such that  $M(x, u)$  accepts in polynomial time. This can also explain why the certificate is required to be in size  $O(T(n))$ .

In mathematical terms:

$$x \in L \Leftrightarrow \exists u \in \{0, 1\}^{O(T(n))} \text{ s.t. } M(x, u) = 1.$$

Then for NP,  $T(n) = \text{poly}(n)$ .

#### Example 2.2.2.

- Independent Set.

$$\text{IND-SET} = \{\langle G, k \rangle : \exists S \subseteq V(G) \text{ s.t. } |S| \geq k \text{ and } \forall u, v \in S, \{u, v\} \notin E(G)\}$$

Independent Set is in NP because we can verify a solution by manually checking the edges in the graph.

- Traveling Salesman Problem (TSP) is about many cities, with  $d_{ij}$  being the distance between city  $i$  and city  $j$ .

The problem is whether there exists a closed circuit that visits each city exactly once, with total distance at most  $k$ .

TSP is in NP because we can calculate the distance of a circuit in polynomial time.

- Linear Programming:  $A\vec{x} \leq \vec{b}$ . Is there a solution  $\vec{x}$ ?

LP is trivially NP, but people later found that LP is also in P.

- Graph Isomorphism: Given two adjacency matrices, do they define the same graph up to permutation of vertices?

Graph Isomorphism is in NP because we can verify a solution by checking if such permutation of vertices gives the same adjacency matrix.

- Factoring.

$$\text{FACTORING} = \{\langle N, k \rangle : N \text{ has a factor } \leq k\}$$

Equivalently, it can also decide  $\{\langle N, L, R \rangle : N \text{ has a prime factor between } L \text{ and } R\}$ .

Factoring is in NP as we can just divide  $N$  by a certificate.

#### Theorem 2.2.1.

$$\text{P} \in \text{NP} \cap \text{coNP}.$$

## Chapter 3

# Proving NP-hard Problems

### 3.1 NP-hard and NP-complete

**Definition 3.1.1** (Karp Reduction).  $L \leq_p L'$  if there's a polynomial-time computable function  $f$  such that for all  $x$ ,  $x \in L \Leftrightarrow f(x) \in L'$ .

**Definition 3.1.2** (NP-hard and NP-complete).  $L$  is **NP-hard** if for all  $L' \in \text{NP}$ ,  $L' \leq_p L$ .  $L$  is **NP-complete** if  $L$  is NP-hard and  $L \in \text{NP}$ .

We can find a NP-complete problem by simply looking at the definitions.

**Problem 3.1.1** (TMSAT).

$$\text{TMSAT} = \{ \langle M, x, 1^n, 1^t \rangle : \exists u \in \{0, 1\}^n \text{ such that } M(x, u) \text{ accepts in } t \text{ steps} \}.$$

**Theorem 3.1.1.** TMSAT is NP-complete.

**Proof.** First of all we see that TMSAT is in NP as we can give out  $u$  and simulate.

Now we want to prove that TMSAT is NP-hard. Let  $L \in \text{NP}$  and let  $M$  be a polynomial-time verifier for  $L$ . Let  $p(n)$  be the polynomial bound on the running time of  $M$ . We can reduce  $L$  to TMSAT by mapping  $x$  to  $\langle M, x, 1^{p(|x|)}, 1^{p(|x|)} \rangle$ . This reduction is clearly polynomial-time computable. Now we have that  $x \in L$  if and only if there exists a  $u$  such that  $M(x, u)$  accepts in  $p(|x|)$  steps, which is equivalent to  $\langle M, x, 1^{p(|x|)}, 1^{p(|x|)} \rangle \in \text{TMSAT}$ . Therefore, TMSAT is NP-hard. ■

This, however, does not give us a very useful NP-complete problem.

### 3.2 Cook-Levin Theorem

Cook-Levin theorem is about an interesting NP-complete problem called SAT.

**Definition 3.2.1** (CNF and SAT). A formula is in **conjunctive normal form** (CNF) if it is a Boolean formula over variables  $u_1, u_2, \dots, u_n$  that has the form

$$\bigwedge_i \left( \bigvee_j v_{ij} \right)$$

where each  $v_{ij}$  is either a variable  $u_k$  or its negation  $\neg u_k$  (also written as  $\overline{u_k}$ ). A formula is **satisfiable** if there exists an assignment of truth values to the variables that makes the formula evaluate to true. The **SAT** problem is the decision problem of determining whether a given CNF formula is satisfiable.

**Theorem 3.2.1 (Cook-Levin Theorem).** SAT is NP-complete.

**Proof.** It suffices to show that SAT is NP-hard.

Let  $L \in \text{NP}$ . By definition, there exists a polynomial-time verifier  $M$  for  $L$ . We want to describe a polynomial-time transformation  $x \rightarrow \varphi_x$  such that

$$\varphi_x \in \text{SAT} \Leftrightarrow \exists u \in \{0, 1\}^{p(|x|)} \text{ such that } M(x, u) = 1$$

We do so by constructing a tableau of the computation of  $M$  on input  $(x, u)$ , where  $u$  is a string of length  $p(|x|)$ . The tableau is a two-dimensional array where each row represents a configuration of the Turing machine at a specific time step, and each column represents a tape cell.

The first row corresponds to the initial configuration. For example, for a starting state  $q_0$  and input tape containing  $x$  (and probably  $u$ ), the first row would look like:

$$\begin{array}{ccccccc} \perp & q_0 & w_1 w_2 \dots w_n & \# \# \dots & \perp \\ & \text{state} & \text{input tape} & \text{blank cells} & \end{array}$$

Checking the first row is easy, as we can hardcode the initial configuration into the formula. Name such a formula  $\varphi_{\text{start}}$ .

Then after some state transitions, the intermediate rows may look like:

$$\begin{array}{ccccccc} \perp & \# \# \dots & y_1 y_2 \dots y_i & q_j & y_{i+1} \dots y_m & \# \# \dots & \perp \\ & \text{blank cells} & \text{tape contents} & \text{state} & \text{tape contents} & \text{blank cells} & \end{array}$$

We can't be sure about the contents of the tape, but we can check that the transition from one row to the next is valid according to the transition function of  $M$ . For example, describing a transition on state  $q_i$  only concerns a  $2 \times 3$  window of the tableau:

$$\begin{array}{ccc} \# & q_i & 0 \\ q_j & \# & 1 \end{array}$$

Then it suffices to encode the transition rules of  $M$  into a windows, and check that every  $2 \times 3$  window of the tableau satisfies one of the transition rules. This can be done by a formula  $\varphi_{\text{move}}$ .

Also, we want to check that all cells are valid (i.e. there're no multiple characters in a cell) We encode such constraints into a formula  $\varphi_{\text{cell}}$ .

Finally, there has to be a  $q_{\text{accept}}$  in the last row of the tableau. Searching for this symbol can be encoded into a formula  $\varphi_{\text{accept}}$ .

Now we construct the final formula as  $\varphi_x = \varphi_{\text{start}} \wedge \varphi_{\text{move}} \wedge \varphi_{\text{cell}} \wedge \varphi_{\text{accept}}$ . This formula is satisfiable if and only if there exists a valid tableau of the computation of  $M$  on input  $(x, u)$  that leads to an accepting configuration, which is equivalent to saying that there exists a  $u$  such that  $M(x, u) = 1$ .

■

# **Appendix**

# Appendix A

## Algebra

**Theorem A.0.1.** For any polynomial  $A \in \mathbb{C}[X]$  of degree  $n - 1$ , any point set  $\{p_0, \dots, p_{n-1}\}$  of size  $n$ ,  $\{(p_i, A(p_i))\}_{i=0}^{n-1}$  determines  $A(x)$  and vice versa.

**Theorem A.0.2** (Fundamental Theorem of Algebra). Every non-constant single-variable polynomial with complex coefficients has at least one complex root.

In math terms, for  $A(x) = \sum_{i=0}^{n-1} a_i x^i$ , either  $\exists x \in \mathbb{C}$  such that  $A(x) = 0$ , or  $a_0 = a_1 = \dots = a_{n-1} = 0$ .

## Appendix B

# Analysis

**Theorem B.0.1 (Master Theorem (simplified)).** Let the recurrence relation be:

$$T(n) = aT\left(\frac{n}{b}\right) + n^d$$

where  $a \geq 1$  is the number of subproblems,  $b > 1$  is the factor by which the problem size is reduced, and  $d \geq 0$ . The asymptotic bound of  $T(n)$  is given by:

1. If  $\log_b a > d$ , then  $T(n) = O(n^{\log_b a})$ .
2. If  $\log_b a = d$ , then  $T(n) = O(n^d \log n)$ .
3. If  $\log_b a < d$ , then  $T(n) = O(n^d)$ .

**Theorem B.0.2 (Master Theorem).** Let  $a \geq 1$  and  $b > 1$  be constants, let  $f(n)$  be a function, and let  $T(n)$  be defined on the nonnegative integers by the recurrence:

$$T(n) = aT(n/b) + f(n)$$

Then  $T(n)$  has the following asymptotic bounds:

- **Case 1:** If  $f(n) = O(n^{\log_b a - \epsilon})$  for some constant  $\epsilon > 0$ , then  $T(n) = \Theta(n^{\log_b a})$ .
- **Case 2:** If  $f(n) = \Theta(n^{\log_b a} \log^k n)$  for some constant  $k \geq 0$ , then  $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$ .
- **Case 3:** If  $f(n) = \Omega(n^{\log_b a + \epsilon})$  for some constant  $\epsilon > 0$ , and if  $af(n/b) \leq cf(n)$  for some constant  $c < 1$  and all sufficiently large  $n$  (Regularity Condition), then  $T(n) = \Theta(f(n))$ .